

SOFTWARE REQUIREMENTS **SPECIFICATION**

FOR

ONLINE FOOD ORDERING SYSTEM

GUIDED BY,
MRS. PINKY M.S
SUBMITTED ON: 16/09/2025

PREPARED BY,
AAYUSHI MICHAEL
ARSHIA P ANIL
M DEVIKA MOHAN
TANIYA SUNNY

Table of Contents

Table of Contents	Error!
Bookmark not defined.	
1. Introduction	2
1.1 Purpose	3
1.2 Document Conventions	4
1.3 Intended Audience and Reading Suggestions	5
1.4 Product Scope	8
1.5 References	9
2. Overall Description	10
2.1 Product Perspective	11
2.2 Product Functions	12
2.3 User Classes and Characteristics	14
2.4 Operating Environment	15
2.5 Design and Implementation Constraint	17
2.6 User Documentation	18
2.7 Assumptions and Dependencies	20
3. External Interface Requirements	21
3.1 User Interfaces	22
3.2 Hardware Interfaces	23
3.3 Software Interfaces	Error!
Bookmark not defined.4	
3.4 Communications Interfaces	Error! Bookmark not defined.5
4. System Features	Error! Bookmark not defined.6
4.1 System Feature 1	Error! Bookmark not defined.7
4.2 System Feature 2 (and so on)	Error! Bookmark not defined.8

5. Other Nonfunctional Requirements30

Appendix A:
Glossary.....31

Appendix B: Analysis
Models.....32

INTRODUCTION

Purpose

This Software Requirements Specification (SRS) document specifies the requirements for the **Online Food Ordering System (OFOS) – Version 1.0**. This system is intended to provide a digital platform that connects customers with local restaurants, enabling users to browse menus, place orders online, and choose delivery or pickup options. The product is a web-based application that can later be extended to mobile platforms in future versions.

*This SRS covers the **complete system** for the Online Food Ordering System (OFOS), not just a single component or subsystem. It includes features for:*

- **Customers** to register, browse restaurants, place orders, make online payments, and track deliveries.
- **Restaurant Owners** to manage menus, update availability, view incoming orders, and change order statuses.
- **Administrators** to manage users, monitor system performance, and ensure smooth operations.

*This document outlines both **functional** and **non-functional requirements** for the initial release (v1.0). Future features such as delivery agent tracking, AI-based recommendations, and mobile app integration are **not included** in this version and will be documented in future revisions.*

Document Conventions

This Software Requirements Specification (SRS) follows the IEEE 830-1998 standard for documenting software requirements. To maintain clarity and uniformity, the following typographical conventions and standards have been applied throughout this document:

1. Fonts and Styles

- **Headings:** Bold and numbered according to section hierarchy (e.g., 1.0, 1.1, 1.2).
- **Body Text:** Times New Roman, 12pt, single spacing for readability.
- **Keywords/Definitions:** Italicized when introduced for the first time.
- **System Components/Modules:** Written in **bold** (e.g., **User Module**, **Admin Panel**).

2. Requirement Identification and Priorities

- Each requirement is uniquely numbered for traceability:
 - Functional Requirements: **FR-1**, **FR-2**, **FR-3**...
 - Non-Functional Requirements: **NFR-1**, **NFR-2**, **NFR-3**...
- Priorities are explicitly stated for each requirement; **higher-level priorities are not assumed to be inherited** by lower-level requirements.
- Priority notation follows:
 - **[MUST]** – Mandatory requirement
 - **[SHOULD]** – Recommended but not essential
 - **[MAY]** – Optional requirement

3. Tables, Figures, and Lists

- Requirements and specifications are presented in tabular format wherever possible.
- Bulleted and numbered lists are used for enumerating features, constraints, or options.
- Figures, diagrams, and tables are numbered sequentially and referenced in the text.

4. HIGHLIGHTING AND NOTES

- Important notes and assumptions are emphasized using *italics* or a separate block.
 - Acronyms and abbreviations are defined upon first use and summarized in the **Glossary** section.

Intended Audience and Reading Suggestions

This document is intended for a variety of stakeholders involved in the development, deployment, and usage of the Online Food Ordering System. Each type of reader may focus on different sections of this Software Requirements Specification (SRS) according to their role and responsibilities:

- **Project Managers:** To understand the overall scope, objectives, and system requirements. They should focus on the Introduction (Section 1.0) and Overall Description (Section 2.0).
- **Developers/Programmers:** To implement the system functionalities. They should focus on Functional Requirements (Section 3.0), External Interface Requirements (Section 3.2), and detailed design constraints.
- **Testers/Quality Assurance Team:** To validate the system against its requirements. They should primarily review Functional and Non-Functional Requirements (Sections 3.0 and 4.0).
- **End Users (Customers and Restaurant Staff):** To gain a high-level understanding of the system's features and workflow. They should focus on the Overview (Sections 1.0 and 2.0).
- **Marketing and Business Analysts:** To align system features with business goals and customer needs. They should review the Introduction (Section 1.0), Overall Description (Section 2.0), and Product Perspective subsections.
- **Documentation Writers/Technical Writers:** To prepare user manuals and guides. They should review the entire SRS, but place particular emphasis on User Interfaces (Section 3.2.1) and Functional Requirements (Section 3.0).

Reading Sequence Suggestions:

1. Begin with **Section 1 (Introduction)** for context, objectives, and definitions.
2. Proceed to **Section 2 (Overall Description)** for understanding the system environment, features, and constraints.
3. Readers requiring implementation details (e.g., developers, testers) should then focus on **Section 3 (Specific Requirements)**.
4. Non-technical readers (e.g., business analysts, marketing staff, users) may stop after Section 2 or selectively review sections relevant to their interests.

This organization ensures that both technical and non-technical stakeholders can effectively use this SRS to meet their respective goals.

Product Scope

The **Online Food Ordering System** is designed to provide a convenient, efficient, and user-friendly platform for customers to browse restaurant menus, place orders, and make secure payments through a web or mobile interface. The system eliminates the need for physical presence in restaurants, reduces waiting times, and enhances the overall dining experience.

This project aims to benefit both **customers** and **restaurants**:

- **For Customers:** The system provides an easy-to-use interface for browsing menus, customizing orders, and making payments. It also enables real-time order tracking and delivery updates.
- **For Restaurants:** The system streamlines order management, reduces human errors in taking orders, and provides sales and performance insights for better business decision-making.

Scope Includes:

- User registration and authentication.
- Browsing menus and selecting food items.
- Order placement, modification, and cancellation.
- Secure online payment integration.
- Real-time order tracking and delivery management.
- Admin panel for restaurant staff to manage menus, prices, and orders.

Out of Scope:

- Physical food delivery (the system supports order and tracking only; actual delivery is outside the scope of software).
- Third-party integrations beyond payment gateways.

The project will ensure reliability, scalability, and data security to handle multiple users simultaneously. By automating the food ordering process, the system will enhance customer satisfaction, increase operational efficiency for restaurants, and create opportunities for business growth.

References

1. *IEEE Recommended Practice for Software Requirements Specifications, IEEE Std 830-1998, IEEE Computer Society, 1998.*Source: <https://ieeexplore.ieee.org/document/720574>
2. *Sommerville, Ian. Software Engineering, 10th Edition, Pearson Education, 2015.ISBN: 978-0133943030.*
3. *Pressman, Roger S. Software Engineering: A Practitioner's Approach, 8th Edition, McGraw-Hill, 2014.ISBN: 978-0078022128.*
4. *ISO/IEC/IEEE 29148:2018, Systems and software engineering — Life cycle processes — Requirements engineering.*Source: <https://www.iso.org/standard/72089.html>
5. *Online Food Ordering Industry Report (2023). Statista – Online Food Delivery Worldwide.*Source: <https://www.statista.com/markets/424/topic/540/online-food-delivery>

6. *TutorialsPoint. Software Requirements Specification (SRS) – Tutorial.Source:*
https://www.tutorialspoint.com/software_engineering/software_requirements_specification.htm
-

Overall Description

Product Perspective

The **Online Food Ordering System** is a **new, self-contained product** designed to automate and simplify the food ordering process for both customers and restaurants. It replaces traditional manual ordering methods (such as telephone-based or in-person orders) with a digital platform accessible via web or mobile applications.

This system is **stand-alone**, but it may interact with:

- **Payment Gateways** for secure online transactions.
- **Notification Services** (e.g., SMS/Email APIs) for sending order confirmations and status updates.
- **Restaurant Management Systems** for menu updates and order tracking.

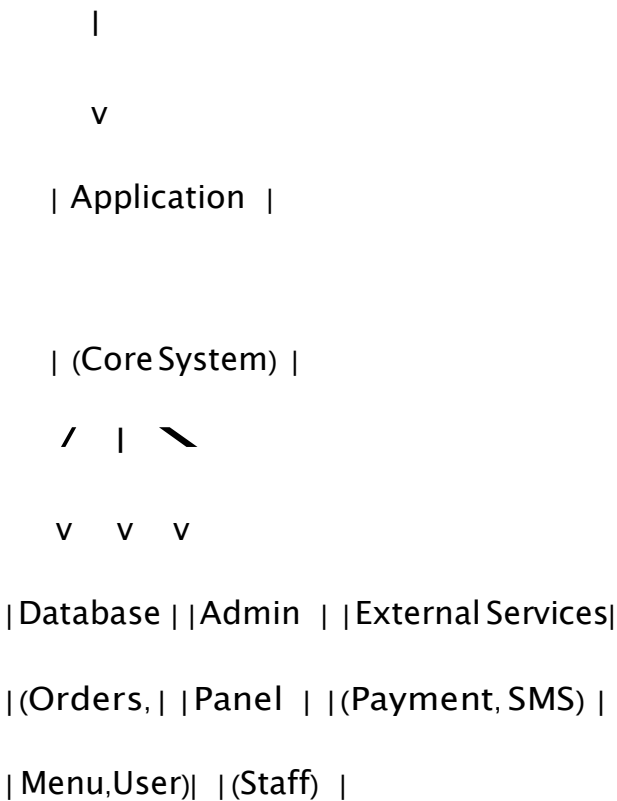
The Online Food Ordering System is not part of a product family but is designed with scalability in mind so that it can be integrated with future services such as delivery partner systems, loyalty programs, or third-party food aggregators.

The major components include:

- **Customer Interface**: For browsing menus, placing orders, and making payments.
- **Restaurant/Staff Interface (Admin Panel)**: For managing menus, processing orders, and updating order status.
- **System Database**: For storing user data, menu details, orders, and transaction records.
- **External Interfaces**: Payment gateway API, notification service API.
- **System Context Diagram**

Here's a simple diagram to illustrate the system perspective:

| Customer/User | <-----> | Web/Mobile App |



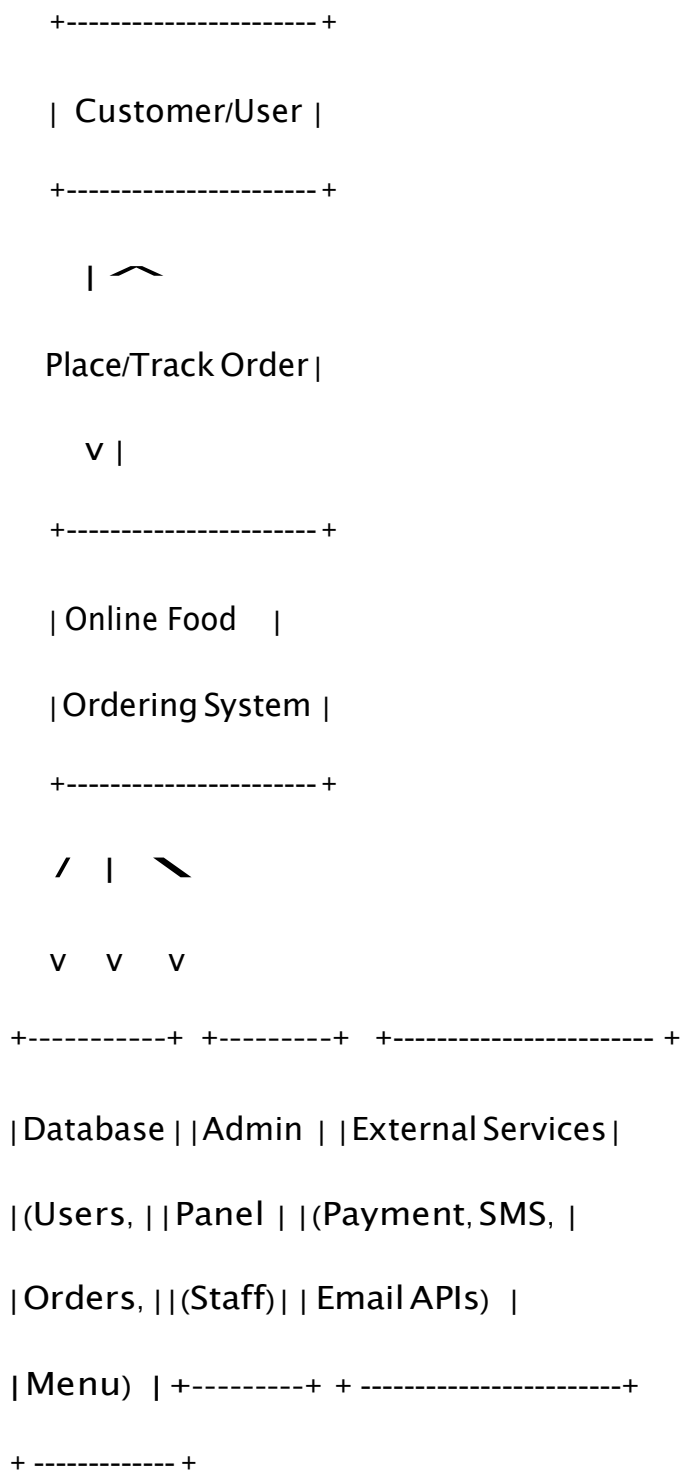
Product Functions

The **Online Food Ordering System** provides the following major functions, grouped according to the type of user:

- 1. **Customer Functions**
 - Register and log in to the system.
 - Browse menus and view food item details.
 - Select food items, customize orders, and add to cart.
 - Place, modify, or cancel orders.
 - Make secure online payments or select cash-on-delivery.
 - Track order status in real-time.
 - Receive notifications (order confirmation, preparation, dispatch, delivery).
- 2. **Restaurant/Staff (Admin Panel) Functions**
 - Manage customer accounts and access control.
 - Add, update, or remove menu items and prices.
 - Receive and process incoming orders.
 - Update order status (e.g., preparing, ready, delivered).
 - Generate reports on sales, popular items, and customer data.
- 3. **System Functions**
 - Store and retrieve customer, menu, and order data.
 - Authenticate users and enforce access control.
 - Integrate with payment gateways for secure transactions.
 - Interface with external notification services (e.g., email, SMS).

- Maintain transaction logs for security and auditing.

• Top-Level Data Flow Diagram (DFD – Context Level)



User Classes and Characteristics

The Online Food Ordering System will be used by multiple categories of users, each with distinct responsibilities, privileges, and technical expertise. The anticipated user classes are as follows:

- **1. Customers (End Users)**
 - **Description:** Individuals who browse menus, place orders, and make payments.
 - **Characteristics:**
 - Varying levels of technical expertise; designed for easy use by non-technical users.
 - Use the system occasionally or frequently depending on their dining habits.
 - Require simple navigation, responsive design, and secure transactions.
 - **Importance: Most important** user class, as they represent the primary target of the system.
- **2. Restaurant Staff (Admins/Operators)**
 - **Description:** Staff members responsible for managing menus, processing customer orders, and updating order status.
 - **Characteristics:**
 - Moderate technical expertise (basic computer or mobile device usage).
 - Frequent users who interact with the system daily.
 - Require quick access to order management tools and reporting features.
 - **Importance: Highly important**, as their efficiency ensures smooth system operation.
- **3. System Administrators**
 - **Description:** Technical staff responsible for maintaining the system, databases, security, and server operations.
 - **Characteristics:**
 - High technical expertise (knowledge of databases, servers, and application security).
 - Occasional users who primarily perform maintenance, updates, and troubleshooting.
 - Access to all system modules with highest privilege level.
 - **Importance: Important**, though less frequent users compared to customers and restaurant staff.
- **4. Delivery Personnel (Optional/Future Integration)**
 - **Description:** Individuals responsible for delivering orders to customers.
 - **Characteristics:**
 - Minimal technical expertise; system access may be limited to order tracking and delivery confirmation.
 - Use system occasionally (for confirming deliveries).
 - **Importance: Less critical** in current scope, but may become important in future system expansions.

Operating Environment

The **Online Food Ordering System** is a web and mobile-based application that will operate in the following environments:

- **1. Client-Side Environment (End Users & Staff)**
 - **Devices:** Desktop computers, laptops, tablets, and smartphones.
 - **Operating Systems:** Windows, macOS, Android, iOS.
 - **Web Browsers:** Google Chrome, Mozilla Firefox, Microsoft Edge, Safari (latest two versions).

- **Mobile Application (if implemented):** Android (v8.0 or higher), iOS (v12 or higher).
- **User Interface:** Responsive design to support multiple screen sizes.
- **2. Server-Side Environment**
- **Operating System:** Linux (Ubuntu/CentOS) or Windows Server.
- **Application Server:** Apache Tomcat / Node.js / or similar web server.
- **Database Management System (DBMS):** MySQL / PostgreSQL / Oracle.
- **Programming Languages:** Java, Python, PHP, or equivalent backend technologies.
- **Frameworks (if applicable):** Spring Boot, Django, or Express.js for backend; React/Angular for frontend.
- **3. Network Environment**
- **Connectivity:** High-speed internet connection (minimum 2 Mbps recommended for users).
- **Protocols:** HTTP/HTTPS for web access, TLS for secure communication.
- **External Integrations:**
 - Payment Gateway API (e.g., PayPal, Razorpay, Stripe).
 - Email/SMS Notification Services.
- **4. Security Environment**
- Secure Sockets Layer (SSL)/Transport Layer Security (TLS) for data encryption.
- Role-based access control for customers, restaurant staff, and administrators.
- Compliance with standard data privacy and security guidelines.

Design and Implementation Constraints

The development of the **Online Food Ordering System** will be subject to the following design and implementation constraints:

- **1. Regulatory and Policy Constraints**
- The system must comply with **data privacy regulations** such as GDPR or equivalent local laws.
- Payment transactions must follow **PCI-DSS standards** to ensure secure handling of cardholder data.
- Restaurant data and customer information must be securely stored and protected from unauthorized access.
- **2. Hardware and Performance Constraints**
- The application must run efficiently on standard consumer devices (desktop and mobile) with limited memory and processing power.
- The server must be capable of handling **at least 500 concurrent users** without significant performance degradation.
- Database queries should be optimized to minimize response times (target: <2 seconds per request under normal load).
- **3. Software and Technology Constraints**
- The system must be developed using **open-source or widely supported technologies** to ensure long-term maintainability.
- The backend must use a **relational database management system** (e.g., MySQL, PostgreSQL).

- The frontend must be implemented as a **responsive web interface** compatible with major browsers and devices.
- All communications must occur via **HTTPS protocol** to ensure secure data transmission.
- 4. **External Interface Constraints**
- Integration is limited to **payment gateways** (e.g., PayPal, Stripe, Razorpay) approved by the institution.
- Notification services (SMS/Email APIs) must follow the restrictions and rate limits of the chosen provider.
- The system should be able to operate independently but allow for **future integration with third-party delivery services**.
- 5. **Security and Maintenance Constraints**
- Role-based access control must be implemented (Customer, Restaurant Staff, System Administrator).
- Passwords must be stored using strong encryption standards (e.g., SHA-256 or bcrypt).
- Logging and auditing must be implemented to detect suspicious activities.
- The design should follow coding conventions and documentation standards to support **future maintenance by other developers**.

User Documentation

The following user documentation components will be delivered along with the **Online Food Ordering System** to assist customers, restaurant staff, and administrators in effectively using the system:

- 1. **User Manuals**
- **Customer User Manual:** Step-by-step guide on account creation, browsing menus, placing and tracking orders, and making payments.
- **Restaurant Staff (Admin Panel) Manual:** Instructions for managing menus, processing orders, updating order status, and generating reports.
- **System Administrator Manual:** Documentation covering system setup, database maintenance, security management, and troubleshooting.
- 2. **Online Help and Tutorials**
- **Built-in Help Section:** Accessible from within the application, providing quick answers to common tasks (FAQs, troubleshooting).
- **Interactive Tutorials:** Walkthroughs for new users on how to register, order food, and use key features.
- **Context-Sensitive Help:** Tooltips and guidance displayed directly in the user interface for specific fields and buttons.
- 3. **Delivery Formats**
- **Digital Documentation:** All manuals will be delivered in **PDF format** for offline access.
- **Web-Based Documentation:** Online help pages integrated into the system for real-time access.
- **Video Tutorials (Optional):** Short instructional videos demonstrating core system functionalities.
- 4. **Documentation Standards**

- Documentation will follow standard **IEEE and ISO/IEC guidelines** for software user documentation.
- Clear, concise language will be used, with screenshots and diagrams to aid comprehension.

Updates to documentation will be provided with each major release of the system.

Assumptions and Dependencies

The development and operation of the **Online Food Ordering System** are based on the following assumptions and dependencies. Any change or failure of these assumptions could impact the requirements and overall performance of the system.

1. **Assumptions**

2. End users (customers and restaurant staff) will have access to a **stable internet connection**.
3. Customers and staff will have access to **modern devices** (smartphones, tablets, or computers) with up-to-date web browsers.
4. Restaurants will allocate **designated staff members** to monitor and manage incoming orders through the admin panel.
5. Payment gateway services will remain **operational, secure, and compliant** with current financial regulations.
6. Users will provide accurate information (e.g., contact details, delivery address) during registration and ordering.
7. The number of concurrent users will not exceed the estimated capacity defined in performance requirements.
8. The system will be deployed on a reliable hosting environment with regular backups and security measures.

9. **Dependencies**

10. **Third-party payment gateways** (e.g., PayPal, Stripe, Razorpay) for secure financial transactions.
11. **External notification services** (SMS, email APIs) for order confirmations and delivery updates.
12. **Database Management System (DBMS)** such as MySQL or PostgreSQL for data storage and retrieval.
13. **Web browsers and mobile platforms** for compatibility (Google Chrome, Firefox, Safari, Android, iOS).
14. **Operating environment** (server hosting, network infrastructure) provided by the deployment platform.
15. Future integration with **delivery service providers** may depend on their APIs and availability.

External Interface Requirements

User Interfaces

The **Online Food Ordering System** will provide an intuitive, user-friendly interface to ensure smooth interaction for customers, restaurant staff, and administrators. The logical characteristics of the user interfaces are as follows:

- 1. **General Design Standards**
- A **consistent layout** will be used across all screens, with standard navigation menus, headers, and footers.
- Standard buttons such as **Home, Back, Help, Logout, and Contact Support** will be available on all screens.
- Error messages will be displayed in **red text** at the top of the screen or near the input field, with clear instructions to resolve the issue.
- Confirmation messages (e.g., “Order Placed Successfully”) will be displayed in **green text** or as a pop-up notification.
- The interface will follow responsive design principles to support **desktop, tablet, and mobile devices**.
- 2. **Customer Interface Screens**
- **Login/Registration Screen**: User authentication with fields for email/phone number and password, along with “Forgot Password” and “Sign Up” options.
- **Home/Menu Screen**: Displays available restaurants and food items, categorized with images, descriptions, and prices.
- **Cart and Checkout Screen**: Allows users to review selected items, apply discounts, and choose payment methods.
- **Order Tracking Screen**: Displays order status (e.g., pending, preparing, out for delivery, delivered).
- **Profile Screen**: Allows users to manage personal details, saved addresses, and order history.
- 3. **Restaurant/Staff (Admin Panel) Screens**
- **Dashboard**: Displays new orders, pending orders, and recent activity.
- **Menu Management Screen**: Allows adding, updating, or removing menu items with images and prices.
- **Order Management Screen**: Provides details of incoming orders with options to update status (e.g., preparing, dispatched).
- **Reports Screen**: Generates reports on sales, most ordered items, and customer activity.

- 4. **System Administrator Screens**
- **Admin Dashboard:** Provides full system overview including user accounts, restaurants, and transactions.
- **User Management Screen:** Allows creating, editing, or deactivating customer/staff accounts.
- **System Settings Screen:** For configuring payment gateways, notification services, and security features.
- 5. **GUI Standards and Constraints**
- The interface will follow modern **Material Design** and **Bootstrap/Responsive Design** standards.
- **Fonts:** Standard web-safe fonts (e.g., Arial, Helvetica, or Roboto).
- **Color Scheme:** Neutral/light background with contrasting colors for buttons and highlights.
- **Accessibility:** Interfaces will comply with **WCAG** 2.0 guidelines for readability and accessibility.

Hardware Interfaces

The **Online Food Ordering System** is primarily a web-based application and therefore has minimal direct interaction with hardware components. However, certain hardware interfaces are necessary for smooth operation:

- 1. **User Devices**
- **Supported Devices:** Desktop computers, laptops, tablets, and smartphones.
- **Interfaces:** The system will interact with the device's input/output hardware (keyboard, mouse, touchscreen, speakers, microphone for voice search, and display screen).
- **Nature of Interaction:** User commands (clicks, taps, or voice input) are captured by the device and processed by the application. Output is displayed on the screen, with optional audio feedback.
- 2. **Server Hardware**
- **Hosting Servers:** The system will run on dedicated or cloud-based servers that manage the database, web services, and application logic.
- **Interfaces:** Interaction occurs through standard protocols (HTTP/HTTPS, TCP/IP).
- **Nature of Interaction:** Servers handle request/response communication with user devices, process transactions, and store order and customer data.
- 3. **Point of Sale (POS) Devices (Optional, for Restaurants)**
- **Supported Devices:** Restaurant POS terminals, receipt printers, and order display systems.
- **Interfaces:** System will connect via APIs or local network protocols to push order details directly to restaurant devices.
- **Nature of Interaction:** Orders received through the system will be transmitted to POS terminals for preparation and billing.
- 4. **Payment Hardware/Devices**
- **Supported Devices:** Card readers, mobile payment terminals (if used by restaurants).
- **Interfaces:** Secure communication protocols (e.g., PCI DSS compliant payment gateways, NFC/UPI interfaces).
- **Nature of Interaction:** The system may send encrypted transaction details to payment devices for processing.

- **5. Network Hardware**
- **Supported Devices:** Routers, modems, Wi-Fi access points, and mobile networks.
- **Interfaces:** Standard communication protocols (TCP/IP, HTTPS).
- **Nature of Interaction:** Ensures data packets are transmitted securely between users, servers, and restaurants.

Software Interfaces

The **Online Food Ordering System** interacts with several software components, both internal and external, to provide complete functionality. The details of these interfaces are provided below:

- **1. Database Interface**
- **Component:** MySQL 8.0 (or PostgreSQL 14).
- **Nature of Communication:** SQL queries executed through JDBC/ODBC or ORM (Hibernate).
- **Inbound Data:** New user registrations, login credentials, food item entries, order requests, payment details.
- **Outbound Data:** User profiles, menu listings, order history, transaction details, delivery status.
- **Shared Data Items:** Customer details, restaurant information, menu database, and payment logs.
- **Constraint:** Data must follow ACID compliance for consistency, and sensitive data (passwords, payments) must be encrypted before storage.
- **2. Operating System Interfaces**
- **Server-Side Supported OS:** Ubuntu Linux 22.04 LTS, Windows Server 2019.
- **Client-Side Supported OS:** Windows 10/11, macOS Ventura, Linux distributions, Android 11+, iOS 15+.
- **Nature of Communication:** System uses HTTP/HTTPS requests processed by Apache Tomcat/Nginx servers.
- **Constraint:** Must maintain cross-platform compatibility for web browsers and mobile devices.
- **3. Web Browsers**
- **Supported Software:** Google Chrome (v100+), Mozilla Firefox (v95+), Microsoft Edge (v100+), Safari (v15+).
- **Inbound Data:** User requests such as login, search queries, order placements, and feedback.
- **Outbound Data:** Dynamic web pages with menus, order status, and transaction confirmations.
- **Constraint:** Must support responsive design (HTML5, CSS3, JavaScript ES6).
- **4. Payment Gateway APIs**
- **Components:** Razorpay v2.0, PayPal REST API v1.3, Stripe API v2022-11-15.
- **Nature of Communication:** Secure HTTPS REST APIs using JSON objects.
- **Inbound Data:** Encrypted payment request (amount, user ID, order ID, payment mode).
- **Outbound Data:** Payment confirmation, transaction ID, refund status.
- **Constraint:** Must comply with PCI-DSS standards and use TLS 1.2+ encryption.
- **5. Notification Services**

- **Components:** Twilio SMS API v2022, Firebase Cloud Messaging v23.0, SMTP (Postfix/SendGrid).
- **Nature of Communication:** REST API calls for SMS/Push notifications, SMTP protocol for emails.
- **Inbound Data:** Notification requests (order updates, promotional offers).
- **Outbound Data:** SMS, email, or push notification messages.
- **Constraint:** All communication must include unique identifiers (Order ID, User ID) for traceability.
- **6. Restaurant POS Integration (Optional)**
- **Components:** Square POS API v2023, Lightspeed POS API v4.0.
- **Nature of Communication:** JSON/XML data exchange over HTTPS.
- **Inbound Data:** New order details (item list, customer info, payment status).
- **Outbound Data:** Order acceptance, preparation status, estimated delivery time.
- **Constraint:** Must handle API response failures gracefully and retry requests when needed.
- **7. Security and Authentication Software**
- **Components:** OAuth 2.0, JWT (JSON Web Tokens), Google reCAPTCHA v3.
- **Nature of Communication:** Middleware validation through API calls and encrypted token passing.
- **Inbound Data:** Login credentials, session tokens, CAPTCHA validation requests.
- **Outbound Data:** Authentication success/failure messages, valid access tokens.
- **Constraint:** All passwords must be hashed using SHA-256 or bcrypt; sessions must expire after inactivity.

Communications Interfaces

The **Online Food Ordering System** requires reliable and secure communication between users, servers, and third-party services. The communication interfaces are defined as follows:

- **1. Web Communication**
- **Protocol:** HTTP/1.1 and HTTPS (TLS 1.2/1.3 for security).
- **Message Format:** JSON will be the primary format for request/response data exchange between the client and server; XML may be used for third-party APIs if required.
- **Purpose:** Handles user interactions via web browsers and mobile applications, including login, browsing menus, placing orders, and tracking status.
- **2. Client–Server Communication**
- **Method:** RESTful APIs over HTTPS.
- **Inbound Messages (Client → Server):** User login requests, food item selections, payment requests, profile updates.
- **Outbound Messages (Server → Client):** Menu data, order confirmations, order status updates, transaction results.
- **Synchronization:** Asynchronous communication supported with AJAX and WebSocket connections for real-time order tracking.
- **3. Payment Gateway Communication**
- **Protocol:** HTTPS with PCI-DSS-compliant encryption.
- **Message Format:** JSON objects containing transaction details (amount, order ID, user ID, payment status).
- **Security:** All payment communication must use SSL certificates with TLS 1.2/1.3 and token-based authentication.
- **4. Email Communication**
- **Protocol:** SMTP (Simple Mail Transfer Protocol).
- **Purpose:** Sends order confirmations, password resets, promotional offers, and administrative notifications.

- **Security:** STARTTLS/SSL encryption for secure email transmission.
- **5. SMS/Push Notification Communication**
- **Protocol:** HTTPS API calls to third-party services (e.g., Twilio, Firebase Cloud Messaging).
- **Message Format:** JSON payload with user identifier, message type, and content.
- **Purpose:** Provides customers with order status updates, delivery alerts, and promotions.
- **6. Restaurant POS / External System Communication (Optional)**
- **Protocol:** REST/GraphQL APIs over HTTPS.
- **Message Format:** JSON/XML for transmitting order details.
- **Purpose:** Syncs incoming orders from the online system to the restaurant's POS or kitchen display system.
- **Security:** Mutual TLS authentication may be required to ensure secure connection between systems.
- **7. Security and Encryption Standards**
- All communication between clients, servers, and third-party systems will be secured with **TLS 1.2/1.3**.
- Sensitive information (passwords, payment details) must be encrypted using **AES-256** before transmission.
- Authentication tokens (JWT/OAuth 2.0) will be used for session management and API access.
- **8. Performance Requirements**
- **Data Transfer Rate:** The system must support at least **1,000 concurrent requests per second** with an average response time below **2 seconds** under normal load.
- **Synchronization:** WebSocket-based communication will provide real-time order status updates.

System Features

The **Online Food Ordering System** provides the following major system features:

- **1. User Registration and Authentication**
- Allows new users (customers, restaurants, and administrators) to create accounts.
- Provides secure login using credentials (email/phone + password).
- Supports password reset and account recovery via email or SMS.
- Uses encryption and authentication tokens (OAuth 2.0/JWT).
- **2. Restaurant and Menu Management**
- Restaurants can register and manage their profiles.
- Restaurant staff can add, edit, or remove menu items with details such as name, price, description, and image.
- The system allows real-time menu updates that are visible to customers.

- **3. Food Ordering and Cart Management**

- Customers can browse menus, search for items, and add them to the shopping cart.
- The cart allows updating quantities, removing items, and applying discount codes.
- Customers can confirm and place orders for delivery or pickup.

- **4. Order Processing and Tracking**

- Restaurants receive incoming orders in their dashboard.
- Staff can update the order status (e.g., confirmed, preparing, out for delivery, delivered).
- Customers can track order progress in real-time through the system or via notifications.

- **5. Payment Processing**

- Supports multiple payment methods: credit/debit cards, UPI, wallets, PayPal, and cash on delivery.
- Integrates with third-party payment gateways for secure transactions.
- Provides transaction confirmation and receipts to both customers and restaurants.

- **6. Delivery Management**

- Assigns orders to delivery staff (manually or automatically).
- Provides delivery agents with route details and customer information.
- Customers can track delivery agents in real-time (if GPS-enabled).

- **7. Notification and Alerts**

- Sends email, SMS, or push notifications to customers about order status, offers, and promotions.
- Alerts restaurants about new or pending orders.
- Provides administrators with system alerts for errors or unusual activity.

- **8. Review and Rating System**

- Customers can rate restaurants and delivery services.
- Restaurants can view feedback for service improvement.
- The system ensures reviews are linked only to verified orders.

- **9. Administration and Reporting**

- Administrators can manage users, restaurants, and transactions.
- Generates reports on sales, order volume, customer activity, and popular items.
- Provides role-based access control to ensure data security.

4.1 User Registration & Authentication (Customer / Restaurant / Admin)

Description & Priority: Provide secure account creation and authentication for Customers, Restaurant Staff, and System Administrators. Priority: High.

Stimulus / Response:

User opens registration → registration form displayed.

User submits valid data → account created and confirmation sent.

User attempts login with valid credentials → access granted and session token issued.

User attempts login with invalid credentials → appropriate error shown and limited retry attempts enforced.

Functional requirements

- FR-UR-01 [MUST] The system shall allow new Customers to register with name, email, phone number, password, and acceptance of terms.
- FR-UR-02 [MUST] The system shall allow Restaurants to create business accounts with business name, contact person, address, menu admin user, and supporting documents (optional).
- FR-UR-03 [MUST] The system shall verify email (via confirmation link) or phone (via OTP) before account activation.
- FR-UR-04 [MUST] The system shall support secure login using email/phone + password and issue a time-limited session token (JWT).
- FR-UR-05 [MUST] The system shall provide “Forgot password” flow via email/OTP and allow secure password reset.
- FR-UR-06 [SHOULD] The system shall support social sign-in (Google) as an optional authentication method.

Acceptance criteria: Registration creates a user row in DB, confirmation sent within 60 sec, login returns JWT with correct scopes.

4.2 Browse Restaurants & Menus

Description & Priority: Customers browse restaurants by category, rating, location and view menus. Priority: High.

Functional requirements

- FR-MN-01 [MUST] The system shall allow customers to search restaurants by name, cuisine, or location.
- FR-MN-02 [MUST] The system shall display restaurant details (name, rating, delivery time, minimum order) and menu categories.
- FR-MN-03 [MUST] The system shall support menu item variants (size, extras) and show item images, description, price, and availability.
- FR-MN-04 [SHOULD] The system shall offer filtering (vegetarian, price range, offers).

Acceptance criteria: Menu pages load in <2s under normal load and correctly show all active items from DB.

4.3 Cart Management & Order Placement

Description & Priority: Allow customers to add items to cart, modify, apply discount codes and place orders. Priority: High.

Functional requirements

- FR-CT-01 [MUST] The system shall provide a persistent cart per authenticated user (and temporary cart for guests stored in session).
- FR-CT-02 [MUST] The system shall allow adding/removing items, changing quantities and selecting variants.
- FR-CT-03 [MUST] The system shall calculate subtotal, taxes, delivery charges, discounts and display final payable amount.
- FR-CT-04 [MUST] The system shall allow order placement with choice of delivery or pickup and selection of saved/new address.
- FR-CT-05 [MUST] The system shall create an Order record with a unique OrderID and initial status Placed.
- FR-CT-06 [MUST] The system shall validate stock/availability at checkout and notify if items become unavailable.

Acceptance criteria: Orders placed persist to DB, payment flow initiated for paid orders, confirmation sent immediately.

4.4 Payment Processing

Description & Priority: Integrate gateway(s) for secure payments and support cash-on-delivery. Priority: High.

Functional requirements

- FR-PAY-01 [MUST] The system shall integrate with at least one payment gateway (e.g., Razorpay/Stripe/PayPal) via REST APIs for card/UPI/wallet payments.
- FR-PAY-02 [MUST] The system shall support Cash-On-Delivery as an alternative payment mode.
- FR-PAY-03 [MUST] The system shall record transaction IDs, gateway responses and payment status in the Order record.
- FR-PAY-04 [MUST] The system shall support refunds (manual via admin or automated for cancelled paid orders).

- FR-PAY-05 [SHOULD] The system shall never store full card details; use tokenization through gateway.

Acceptance criteria: Successful payments update Order status to Paid, unsuccessful payments leave order in Payment Failed and notify user.

4.5 Order Processing & Status Tracking

Description & Priority: Allow restaurants to process orders and customers to track status. Priority: High.

Functional requirements

- FR-OP-01 [MUST] The system shall notify the restaurant of new orders (dashboard + push/SMS optionally).
- FR-OP-02 [MUST] The restaurant staff shall be able to update order statuses: Accepted, Preparing, Ready, Out for delivery, Delivered, Cancelled.
- FR-OP-03 [MUST] The system shall allow estimated delivery time updates and display them to customers.
- FR-OP-04 [MUST] The system shall allow customers to view real-time order status and history.
- FR-OP-05 [SHOULD] For future scope, the system shall support delivery agent assignment and live location tracking.

Acceptance criteria: Status updates reflected in customer UI within 5 seconds.

4.6 Notifications

Description & Priority: Notify users via email, SMS, and push about order events and offers. Priority: Medium.

Functional requirements

- FR-NF-01 [MUST] The system shall send order confirmations by email/SMS on order placement.
- FR-NF-02 [MUST] The system shall send status update notifications at each major order state change.
- FR-NF-03 [SHOULD] The system shall support scheduled promotional emails and in-app messages.

Acceptance criteria: Notification delivery attempts logged; critical notifications reach user within 60-120s.

4.7 Restaurant Admin Panel (Menu & Order Management)

Description & Priority: Restaurant staff manage menu, availability and orders. Priority: High.

Functional requirements

- FR-RA-01 [MUST] The system shall provide a secure restaurant dashboard to add/edit/delete menu items, set prices, and manage availability (per item/time).
- FR-RA-02 [MUST] The system shall present incoming orders in real-time with accept/reject actions and estimated prep time.
- FR-RA-03 [MUST] The system shall produce daily/weekly sales and item reports
- FR-RA-04 [SHOULD] The system shall support printing order slips for kitchen (via POS or PDF).

Acceptance criteria: Admin actions saved and reflected to customers immediately.

4.8 Reviews & Ratings

Description & Priority: Customers rate restaurants and leave feedback. Priority: Medium.

Functional requirements

- FR-RV-01 [MUST] Customers who completed orders may submit a rating (1-5) and optional review text.
- FR-RV-02 [SHOULD] The system shall allow restaurant staff to respond to reviews.
- FR-RV-03 [MUST] Reviews shall be linked only to verified orders to prevent fake reviews.

Acceptance criteria: Reviews appear on restaurant page after moderation rules are passed.

4.9 Administration & Reporting

Description & Priority: System admin manages users, restaurants and monitors system health. Priority: High.

Functional requirements

- FR-ADM-01 [MUST] Admin shall be able to create/disable user accounts, reset passwords, and manage restaurant approvals.
- FR-ADM-02 [MUST] The system shall maintain audit logs for critical actions (user creation, role changes, refunds).

- FR-ADM-03 [MUST] The system shall provide reports: total orders, revenue, active users, disputes.
- FR-ADM-04 [SHOULD] The system shall provide dashboards for KPI visualization (orders/day, revenue/day).

Acceptance criteria: Admin can perform actions and retrieve reports within UI; audit logs queryable.

4.10 Search & Promotion (Coupons)

Description & Priority: Coupon codes, promotions. Priority: Medium.

Functional requirements

- FR-PM-01 [MUST] The system shall support coupon codes with type (fixed/percentage), validity window, per-user limits, and minimum order value rules.
- FR-PM-02 [SHOULD] Coupons may be targeted (e.g., new user only, restaurant-specific).

Acceptance criteria: Coupon logic correctly applied to final payment calculation.

4.11 Security & Privacy

Description & Priority: Ensure secure data storage, role-based access control and privacy compliance. Priority: High.

Functional requirements

- FR-SC-01 [MUST] The system shall implement RBAC with roles: Customer, RestaurantStaff, Admin, Delivery Agent.
- FR-SC-02 [MUST] Passwords must be hashed using a strong algorithm (bcrypt or equivalent) before storage.
- FR-SC-03 [MUST] Sensitive PII and payment information shall be stored in encrypted form (or not stored if tokenized by gateway).
- FR-SC-04 [MUST] All data access must be logged (who, when, what).

Acceptance criteria: Security test shows no plaintext password storage, access control enforc

5.1 Performance Requirements

- NFR-PERF-01 [MUST] The system shall handle at least 500 concurrent users for the initial deployment and be scalable to 2000 concurrent users with horizontal scaling.
- NFR-PERF-02 [MUST] Page response times for common pages (home, menu) shall be < 2 seconds under normal load.
- NFR-PERF-03 [SHOULD] Checkout transaction flow shall complete in < 5 seconds (excluding external payment processing time).
- NFR-PERF-04 [SHOULD] System shall support 1000 order placements per hour during peak.

5.2 Availability & Reliability

- NFR-AV-01 [MUST] The system shall be available 99.5% (monthly) excluding scheduled maintenance.
- NFR-AV-02 [MUST] Automatic retry logic for transient failures (e.g., notification API), with exponential backoff up to 3 retries.

5.3 Security Requirements

- NFR-SEC-01 [MUST] All network communication shall use HTTPS (TLS 1.2+).
 - NFR-SEC-02 [MUST] Comply with relevant payment security standards (PCI-DSS) for handling card payments.
 - NFR-SEC-03 [MUST] Implement input validation and parameterized queries to prevent SQL injection.
 - NFR-SEC-04 [MUST] Implement rate limiting on authentication endpoints to prevent brute-force attacks.
-
- NFR-UX-01 [SHOULD] The UI shall follow responsive design so it is usable on phones, tablets and desktops.
 - NFR-UX-02 [SHOULD] Conform to WCAG 2.0 AA accessibility for core flows (registration, checkout).
 - NFR-UX-03 [MUST] Common workflows (register -> order) should require no more than 6 user actions for an experienced user.

5.4 Maintainability & Supportability

- NFR-MNT-01 [SHOULD] Codebase shall maintain modular services with automated unit

and integration tests achieving >60% code coverage initially.

- NFR-MNT-02 [MUST] Provide logging and monitoring (error, performance) and alerting dashboards for ops team.

5.5 Scalability

- NFR-SCL-01 [SHOULD] Scale horizontally (stateless app servers behind load balancer, DB read replicas) to support growth.
- NFR-SCL-02 [SHOULD] Use caching (Redis or Memcached) for hot reads (restaurant lists, menus) to reduce DB load.

5.6 Localization & Internationalization

- NFR-L10N-01 [MUST] Support UTF-8 for multi-language content; allow locale-based date and currency formatting.
- NFR-L10N-02 [MAY] Allow content translation using external resource files for future multi-language support.

5.7 Legal & Compliance

- NFR-LGL-01 [MUST] Store and handle personal data in compliance with applicable privacy laws (e.g., GDPR-like practices for users).
- NFR-LGL-02 [MUST] Payment flows must comply with regional financial regulations and tax requirements.

Other Requirements

6.1 Database Requirements

Use a relational RDBMS (MySQL/PostgreSQL). Tables shall be normalized to at least 3NF for core transactional data.

Regular automated backups: daily full backups and hourly incremental backups retained for 30 days.

6.2 Deployment & Operating Environment

Primary deployment on cloud VPS or managed cloud (AWS/GCP/Azure) running Linux (Ubuntu 20.04+).

Use containerization (Docker) and orchestration (optional: Kubernetes) for production deployments.

CI/CD pipeline using GitHub Actions / GitLab CI for automated build/test/deploy.

6.3 Audit & Logging

All critical actions (refunds, user role changes, admin logs) logged with timestamp, actor and result.

Maintain logs for at least 180 days (rotated/archived).

6.4 Backup & Recovery

RTO (Recovery Time Objective): 4 hours for core services.

RPO (Recovery Point Objective): 1 hour for transactional data.

6.5 Data Retention & Privacy

Personal data retention by default: active accounts retained; after account deletion, personal data anonymized 30 days after deletion unless required for legal reasons.

Provide users with access to their data and a flow for deletion/export upon request.

6.6 Third-party Integrations

Payment Gateway (Razorpay/Stripe/PayPal)

Notification Service (Twilio or local SMS provider), Email (SendGrid/Postfix)

Optional POS integrations (future scope)

Appendix A: Glossary

Order ID: Unique system identifier for an order.

Customer: End user placing food orders.

Restaurant Staff: Authorized personnel who manage a restaurant's menu and orders.

Admin: System administrator with elevated privileges.

JWT: JSON Web Token used for session authentication.

RPO/RTO: Recovery Point/Time Objective.

Appendix B: Analysis Models

B.1 Context-level Data Flow Diagram (Customer ↔ System ↔ Restaurant ↔ Payment Gateway)

B.2 Level-1 DFD (Cart, Checkout, Order Processing)

B.3 ER Diagram (Users, Restaurants, Menus, Items, Orders, Transactions, Reviews)

B.4 Sequence Diagrams: (1) Place Order and Payment, (2) Restaurant Accepts Order

Appendix C: To Be Determined List

1. TBD-001: Final choice of payment gateway and exact API version.
2. TBD-002: Whether to support multi-restaurant consolidation promotions.
3. TBD-003: Exact SLA figures and hosting provider for production.
4. TBD-004: Delivery agent app scope and APIs (out of scope in v1.0).
5. TBD-005: Detailed acceptance test cases and test data sets.

